

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	19-February-2026	Section	6A , 6B
	Exam Type:	Quiz 2 - Chapter 2	Page(s):	2

Student Name

Roll No.

Section:

Q1. A client host in Lahore needs to resolve the IP address for gaia.cs.umass.edu. Assume the following average Round Trip Times (RTT): **[2 + 2 + 1 = 5 Marks][CLO 2]**

- Local DNS Server to Client: 5 ms
 - Local DNS to Root Server: 100 ms
 - Local DNS to .edu TLD Server: 80 ms
 - Local DNS to Authoritative Server: 120 ms
- a) Calculate the total time elapsed from the moment the client sends the DNS query until the IP address is received, assuming an iterative query process and that no records are currently cached.
- b) Suppose the Local DNS server has already cached the IP address of the .edu TLD server from a previous request. Calculate the new total resolution time.
- c) Once the IP is resolved, the client initiates a non-persistent HTTP connection to fetch a base HTML file. If the RTT between the client and the target web server is 150 ms, how much total time passes from the initial DNS query (Part A) until the TCP connection for the HTTP request is fully established?

Q2: A client requests a base HTML file of size 5 KB. This HTML file references **4** very small embedded images (negligible size).

- RTT (Round Trip Time) between client and server is 100 ms.
- Transmission time for the base HTML file is 5 ms.
- Transmission time for the embedded images is negligible (0 ms).
- Assume the TCP connection setup takes 1 RTT.

Calculate the total time elapsed from when the client clicks the link until the base object and all 4 embedded objects are received for the following scenarios:[**4 + 3 + 3 = 10 Marks**][CLO 2]

a) Non-persistent HTTP

b) Persistent HTTP (without pipelining):

c) Persistent HTTP (with pipelining):